

CYBERNETICS

CONNECTIONISM

DEEP LEARNING

Foundations

Vision

Language

Compute

Reasoning
& Agents

Biology

MCP Neuron / Hebbian Learning
1943-49Perceptron
1958Backpropagation
1986Universal Approximation
1989ReLU
2011Dropout / Adam
2014BatchNorm
2015Cognitron / Neocognitron
1975-80Backprop CNN / LeNet-5
1989-98ImageNet
2009VGG / Inception
2014ResNet / U-Net
2015Mask R-CNN
2017DeepLabV3+
2018Vision Transformer
2020Segment Anything
2023SAM 2 / SAM 3
2024-25SegFormer
2021Mask2Former
2022RNN / LSTM
1990-97Word2Vec
2013Neural Language Model
2003Seq2Seq / Neural Attention
2014-15GPU Deep Learning
2009Transformer
2017GPT-1 / BERT
2018GPT-2
2019GPT-3
2020CLIP
2021Megatron-LM
ZeRO / 3D Parallelism
2019-21FSDP
2023Scaling Laws
2020Chinchilla
2022PPO / RLHF
2017InstructGPT / CoT / ReAct
2022Eubiota
2026DPO / OthelloGPT / ToT / Toolformer
2023DeepSeekMath / GRPO
Talk Like a Graph / G-Retriever
2024DeepSeek-R1
2025MENTOR-IA
2026Seed Counting / Synthetic Plants
2018SNP2Vec / SNVformer
2022Deep Phenotyping + GWAS
2019GPformer / DNABERT-2
2023DNABERT
2021ScGPT / Caduceus
2024Few-Shot Leaf Traits
2023Nucleotide Transformer
GEFormer / ESM3
2025

1940 1950 1960 1970 1980 1990 2000 2010

2020

2025